

System Requirements Specification (SRS)

Rider Payroll & Accounting ERP

1. SYSTEM OVERVIEW

This ERP manages riders, payroll, fines, bikes, expenses, journals, and cashflow for rider outsourcing companies. Built on strict double-entry accounting principles (Journals)

Users:

1. PRO (limited access)
2. Accountant (Full/admin level access, Accountant is the admin)

STEP 1 — CORE ACCOUNTING PHILOSOPHY (NON-NEGOTIABLE RULES)

This step defines rules that **everything else depends on**.
If these are wrong, the whole system collapses.

1.1 Journal Is the Absolute Root

Rule 1.1.1 — Journal Mandatory

- **NO transaction can exist without a Journal**
 - Expense
 - Salary
 - Loan
 - Fine
 - Drawer balance correction
 - Capital injection
 - Profit withdrawal
- Even **manual balance corrections** MUST create a journal.

✓ Confirmed: **YES**

Rule 1.1.2 — Journal Immutability

- Once a Journal is **POSTED**:
 - ✗ Cannot be edited
 - ✗ Cannot be deleted

- ✓ Can only be **reversed**
- Reversal creates:
 - A **new journal**
 - With **equal and opposite ledger entries**
 - Linked to original journal via `reversal_of_journal_id`

✓ Confirmed: **YES (only reversed)**

Rule 1.1.3 — Double-Entry Enforcement

For **every journal**:

- Sum of **Debit** = **Sum of Credit**
- System **hard-blocks save** if imbalance exists

⚠ No override, even for admin.

1.2 Ledger vs Expense (Final Terminology Lock)

Ledger (What it IS)

- Ledger is **NOT a module for creating data**
- Ledger is a **read-only transactional history**
- It answers:
 - Who owes how much?
 - Who was paid how much?
 - When did money move?

Ledger entries are **generated automatically** from journals.

Expense Module (What it IS)

- Expense module is a **journal creation interface**
- Accountant creates:
 - Expense journals
 - Loan journals
 - Salary journals
 - Fine journals
 - Manual adjustment journals
- Expense Categories:

- Configurable
- Drive **default behavior**, not accounting truth

✓ This matches your corrected understanding.

1.3 Receivable / Payable — Mandatory Decision Point

Rule 1.3.1 — Mandatory Flag

While creating **ANY journal**, system **MUST** require:

- Is this:
 - Receivable?
 - Payable?
 - Pure expense?

✓ Confirmed: **Mandatory**

Rule 1.3.2 — What Receivable Means

If marked **Receivable**:

- Journal **MUST** include:
 - Entity Type (Rider / Vendor / Supplier)
 - Entity ID
 - Amount receivable
- This amount:
 - Appears in that entity's ledger
 - Appears in rider's payslip (if rider)

Example:

Visa cost = 2,000

Charged to rider = 5,000

→ Receivable = 5,000 (NOT 2,000)

Rule 1.3.3 — What Payable Means

If marked **Payable**:

- Creates liability in ledger
 - Must be cleared later via payment journal
-

1.4 Drawers — Absolute Money Containers

Rule 1.4.1 — Drawer Is a Ledger Entity

- Drawer is NOT a balance field
- Drawer balance = sum of ledger entries affecting it

Drawers:

- Bank
 - Cash
 - Noqodi (Govt card)
 - Others later
-

Rule 1.4.2 — Manual Drawer Adjustments

Allowed via:

- Journal Type: `Manual Adjustment`
- Mandatory reason text
- Mandatory source drawer

Use cases:

- Capital injection
- Profit withdrawal
- Bank mismatch correction

✓ Confirmed: **Yes**

1.5 Approval Flow (PRO vs Accountant)

Rule 1.5.1 — PRO Journals

Rule 1.5.1 — Actor-Based Transaction Separation (UPDATED)

This rule is **in addition to** approval flow, not instead of it.

1.5.1.a — Actor Attribution (Mandatory)

Every journal **MUST** store:

- created_by_user_id
- created_by_role (PRO / Accountant / System)
- approved_by_user_id (nullable)
- approved_at

This is **immutable metadata**.

1.5.1.b — Separate Histories (Reporting Rule)

System **MUST** allow owner/admin to:

- View **PRO-created journals only**
- View **Accountant-created journals only**
- Compare:
 - Total amount
 - Category-wise spending
 - Receivable vs pure expense
- Time-range filterable

⚠ Even if accountant approves a PRO journal:

- It is still **counted under PRO's initiated expenses**
-

Rule 1.5.2 — Accountant Approval

- Accountant can:
 - Approve
 - Reject
 - Modify receivable/payable amounts
 - On approval:
 - Journal becomes POSTED
 - Ledger entries generated
 - Drawer affected
-

Rule 1.5.3 — Templates (Critical)

- Accountant must be able to:
 - Save **Journal Templates**
 - Auto-fill:
 - Accounts
 - Drawer
 - Category
 - VAT
- Only variable input:
 - Amount
 - Entity (Rider)
 - Receivable override

✓ This is essential for usability — correctly identified.

1.6 Hard System Blocks (No Soft Warnings)

The system **MUST BLOCK** (not warn):

- ✗ Posting journal without receivable/payable decision
- ✗ Posting unbalanced journal
- ✗ Editing posted journal
- ✗ Payroll finalization with Rider ID mismatch
- ✗ Salary posting if rider alias unresolved

STEP 2 — RIDER IDENTITY, ALIASES & COMPANY SWITCHING (CRITICAL)

This is one of the **most important parts** of your system.
If this is wrong, **payroll breaks silently**.

2.1 Rider Identity — Core Principle

Rule 2.1.1 — Internal Rider ID (Primary Key)

- System generates a **permanent internal rider ID**
- This ID:
 - NEVER changes

- Is NEVER visible externally
- Is used for:
 - Ledger
 - Payroll
 - Fines
 - Expenses
 - Historical records

Example:

```
internal_rider_id = RDR-UUID-123
```

Rule 2.1.2 — External IDs Are NOT Primary Keys

The following are **NOT** primary identifiers:

- Talabat ID
- Keeta ID
- Rider Name
- Bike Plate
- Phone number

They are **aliases only**.

Rule 2.1.3 — Emirates ID (Long-Term Anchor)

- Emirates ID is:
 - Mandatory
 - Unique
- Used ONLY for:
 - Rare audits
 - Long-term record recovery
- ✗ Not used in daily searches
- ✗ Not used in payroll matching

This solves:

- Name duplication
 - Rider ID changes
 - Multi-year audits
-

2.2 Rider Aliases (This Solves 90% of Confusion)

Rule 2.2.1 — Rider Alias Table

Each rider can have **multiple aliases**.

Each alias contains:

- `internal_rider_id`
 - `platform` (Talabat / Keeta / Bykea / etc)
 - `external_rider_id` (the ID used in salary sheets)
 - `valid_from_date`
 - `valid_to_date` (nullable)
 - `status` (Active / Inactive)
-

Rule 2.2.2 — One Active Alias Per Platform

For a given rider + platform:

- Only ONE alias can be active at a time

System MUST enforce this.

2.3 Company / Platform Switching Logic

Scenario

Rider A:

- Jan–June → Talabat → ID = TLB123
 - July onwards → Keeta → ID = KTA456
-

Rule 2.3.1 — Who Can Change It

- ONLY Accountant can:
 - Add
 - Activate
 - Deactivate rider aliases
- PRO:
 - Can view

- Cannot edit

Rule 2.3.2 — How Change Happens

When rider changes company:

1. Accountant:
 - Deactivates old alias (sets `valid_to_date`)
 - Creates new alias with new platform + ID
2. Internal rider ID remains same
3. Ledger remains continuous

Rule 2.3.3 — Salary Sheet Matching (System Generated Journal)

When salary sheet is uploaded:

- System matches:
 - Platform
 - External Rider ID
 - Date range
- Finds correct **internal rider ID**
- Generates **system journal**

✓ You already approved system-generated journals.

2.4 HARD BLOCKS (No Mercy Here)

Rule 2.4.1 — Rider ID Mismatch

If salary sheet contains:

- Rider ID not found
- OR alias exists but date range mismatch

→ System:

- ✗ Blocks payroll finalization
- ▶ Flags issue in **Action Needed**
- ✗ Does NOT create journal

You already confirmed this. Locked.

Rule 2.4.2 — Duplicate Alias

If accountant tries to:

- Assign same external rider ID
- To two internal riders

→ System BLOCKS save.

2.5 Historical Accuracy (Very Important)

Rule 2.5.1 — Past Payslips

When extracting salary of:

- 1 year ago
- 2 years ago

System:

- Uses alias valid **during that month**
- NOT current alias

This guarantees:

- Audit correctness
 - Legal defensibility
-

2.6 Rider Status (Connected to Payroll)

Each rider has:

- Active
- Vacation
- Retired

Effects:

- Retired:
 - No new salary journals allowed
 - Ledger remains readable
- Vacation:
 - Salary allowed
 - Marked separately

STEP 3 — PAYROLL ENGINE (END-TO-END)

Payroll is **NOT just salary**.
It is a **system-generated accounting pipeline**.

3.1 Core Payroll Philosophy

Rule 3.1.1 — Payroll Is System-Generated

- Payroll journals are **ALWAYS system-generated**
- Accountant cannot manually post salary journals
- Manual input = configuration only

✓ Already approved.

Rule 3.1.2 — One Payslip Per Rider Per Month Per Platform

For a given:

- Rider (internal ID)
- Platform (Talabat / Keeta / etc)
- Month

→ Exactly **ONE final payslip** can exist.

System **MUST** block duplicates.

3.2 Salary Sheet Ingestion (Parent Company)

Rule 3.2.1 — Salary Sheet Upload

Accountant uploads:

- Platform-specific salary sheet
- Month + Year mandatory
- Platform mandatory

Sheet contains:

- External Rider ID
 - Gross Salary
 - Parent deductions (if any)
 - Net paid by platform (optional)
-

Rule 3.2.2 — Alias Resolution (Hard Gate)

For EACH row:

System tries:

1. Match External Rider ID
2. Match Platform
3. Match Date within alias validity

If FAIL:

- ✕ Block payroll finalization
- ► Flag “Rider Alias Mismatch”
- Row stays unresolved

No bypass.

3.3 Pre-Payslip State (Staging Area)

Before finalization, system creates a **Draft Payslip**.

Draft Payslip contains:

- Gross salary (from platform)
- Platform deductions (read-only)

- Company deductions (editable)
- Outstanding balances snapshot

⚠ No ledger entries yet.

3.4 Company Deductions (Your Real Logic)

Rule 3.4.1 — Deduction Sources

Company deductions can come from:

- Visa receivable
- Loans / advances
- Fines
- Toll (Salik)
- SIM charges
- Any future receivable category

All deductions are:

- Backed by ledger receivables
 - Never invented manually
-

Rule 3.4.2 — Manual Deduction Control

Accountant:

- Sees **total outstanding per category**
- Inputs:
 - How much to deduct THIS month
- System shows:
 - Remaining balance AFTER deduction (live)

This avoids tab switching — as you requested.

Rule 3.4.3 — Over-Deduction Block

System MUST block:

- Deduction > outstanding receivable

- Deduction > salary net available

No negative surprises.

3.5 Deduction Allocation Strategy

Rule 3.5.1 — Allocation Mode

System supports:

- Uniform distribution
- Priority-based distribution

BUT:

☐ **Decision:**

- Allocation is **manual for now**
- Automation can be added later

System MUST:

- Track category-wise reduction
-

3.6 Final Payslip Generation (Point of No Return)

Rule 3.6.1 — Finalization Lock

When accountant clicks **Finalize Payslip**:

- System performs ALL validations
- Generates salary journal
- Generates deduction journals
- Posts to ledger
- Deducts from drawer

After this:

- Payslip becomes READ-ONLY
-

Rule 3.6.2 — Journal Structure (Critical)

For EACH rider finalization:

Journal A — Salary Payable

- Debit: Platform Receivable / Income
- Credit: Rider Salary Payable

Journal B — Company Deductions

- Debit: Rider Payable
- Credit: Each receivable category

Journal C — Net Payment

- Debit: Rider Payable
- Credit: Bank Drawer

All linked via:

- payroll_batch_id
- rider_id
- month

3.7 Drawer Impact

Rule 3.7.1 — Drawer Deduction Timing

- Drawer is affected ONLY at:
 - Net payment journal
- Not at draft stage
- Not at sheet upload

This keeps cash accurate.

3.8 Rider Status Rules

Active

- Normal payroll

Vacation

- Payroll allowed
- Flagged in payslip

Retired

- ✗ Payroll blocked
- ✗ New deductions blocked
- Ledger still visible

3.9 Payslip Content (What Rider Sees)

Payslip MUST show:

- Platform name
- Gross salary
- Platform deductions
- Company deductions (category-wise)
- Opening outstanding
- Deducted this month
- Closing outstanding
- Net paid

This prevents disputes.

3.10 HARD BLOCKS (NON-NEGOTIABLE)

System BLOCKS if:

- Rider alias unresolved
- Duplicate payslip exists
- Outstanding mismatch
- Drawer insufficient funds
- Rider status = Retired

STEP 4 — FINES, BIKES & PLATE MAPPING

4.1 Core Principle (Lock This First)

Rule 4.1.1 — Fines Belong to RIDERS, Not Bikes

- Fine is **financially charged to the rider**
- Bike / plate is only a **routing mechanism**
- Final liability always lands on:
 - `internal_rider_id`

✓ You explicitly confirmed this earlier.

4.2 Bike & Plate Identity

Rule 4.2.1 — Bike Is a First-Class Entity

Each bike has:

- Internal bike ID (permanent)
 - Plate number (unique)
 - Status (Active / Maintenance / Sold)
-

Rule 4.2.2 — Plate Number Uniqueness

- Plate number must be **globally unique**
 - System blocks duplicate plates
-

4.3 Bike ↔ Rider Assignment (Time-Based)

This is **critical**.

Rule 4.3.1 — Assignment Table

Bike assignments are stored with:

- Bike ID
- Rider internal ID
- Assigned From (date + time)
- Assigned To (date + time, nullable)

Rule 4.3.2 — No Overlapping Assignments

- One bike → one rider at a time
- System blocks overlapping ranges

Rule 4.3.3 — Who Can Assign

- ONLY Accountant can:
 - Assign
 - Unassign
 - Modify assignment dates
- PRO:
 - View only

Reason: financial liability.

4.4 Fine Ingestion

Rule 4.4.1 — Fine Entry Sources

Fines can enter system via:

- Manual entry
- Excel upload
- API (future)

Mandatory fields:

- Plate number
- Fine amount
- Fine date
- Fine reference number

Rule 4.4.2 — Plate-Based Resolution

On fine creation:

1. System finds bike by plate
2. Finds rider assigned **at fine date/time**

3. Links fine to that rider

4.5 Unassigned / Ambiguous Fines

Rule 4.5.1 — No Assignment Found

If:

- Plate exists
- BUT no rider assigned at fine time

Then:

- Fine status = UNASSIGNED
- ✕ No journal created
- ▶ Appears in Action Needed

Rule 4.5.2 — Multiple Matches (Should Never Happen)

If overlapping assignments exist (data corruption):

- System BLOCKS fine posting
- Requires admin resolution

4.6 Fine Journal Creation

Rule 4.6.1 — Fine Is a Receivable

Once rider is resolved:

- System generates journal:
 - Debit: Rider Receivable
 - Credit: Fine Expense / Toll Clearing Account
 - Category: Fine / Salik
 - Receivable = mandatory
-

Rule 4.6.2 — Drawer Impact Timing

- Drawer is impacted ONLY when:
 - Fine is paid externally (bank deduction)
 - OR rider salary deduction occurs

This prevents double counting.

4.7 Toll (Salik) Special Rule

Rule 4.7.1 — Toll Logic

- Toll is:
 - Paid by platform
 - Deducted from company account
- Rider must reimburse company

Thus:

- Always receivable from rider
- Mandatory deduction path in payroll

This matches your clarification.

4.8 Fine Deductions in Payroll

Rule 4.8.1 — Fine Visibility

In payslip:

- Fine opening balance
 - Fine deducted this month
 - Fine remaining
-

Rule 4.8.2 — Partial Deductions Allowed

- Accountant manually decides amount
- System enforces:
 - No over-deduction

4.9 Fine Lifecycle

Fine statuses:

- Pending (ingested)
- Unassigned
- Assigned
- Partially recovered
- Fully recovered

Status auto-updates from ledger activity.

4.10 HARD BLOCKS

System BLOCKS:

- Payroll finalization if unassigned fines exist for that rider
- Deleting bikes with financial history
- Editing assignment dates if fines exist in that period

5. RIDER IDENTITY & ALIASES

Internal rider ID permanent. External IDs via aliases. Auto-close previous alias. Platform overlap allowed. Emirates ID mandatory.

STEP 5 — DRAWER & CASHFLOW CONTROL

5.1 Core Drawer Philosophy

Rule 5.1.1 — Drawer Is Not Editable Balance

- Drawer **does not store balance as truth**
- Balance = sum of all posted ledger entries
- `current_balance` is **derived / cached only**

No manual typing of balance.

5.2 Drawer Types

System **MUST** support multiple drawers:

Drawer Type	Examples
Bank	Emirates NBD, Mashreq
Cash	Office Cash
Govt Wallet	Noqodi
Virtual	Clearing / Suspense

Each drawer has:

- ID
 - Name
 - Type
 - Currency
-

5.3 Money Movement Rules

Rule 5.3.1 — All Movement via Journal

IN or OUT:

- Salary payment
- Visa payment
- Capital injection
- Profit withdrawal
- Adjustment

→ **Journal required** (already locked in Step 1)

Rule 5.3.2 — Drawer Direction Logic

- Debit Drawer → Money IN
- Credit Drawer → Money OUT

System **MUST** auto-label direction.

5.4 Manual Adjustments (Critical but Dangerous)

Rule 5.4.1 — Allowed but Controlled

Manual drawer adjustments allowed ONLY via:

- Journal Type = Manual Adjustment
- Mandatory:
 - Reason
 - Reference
 - Approved by accountant

Use cases:

- Capital injection
- Profit extraction
- Bank mismatch correction

Rule 5.4.2 — Adjustment Audit

Manual adjustments:

- Flagged in reports
- Counted separately
- Cannot be hidden

Owner visibility guaranteed.

5.5 Drawer Segregation (Important)

Rule 5.5.1 — No Cross-Drawer Leakage

A journal MUST specify:

- Source drawer
- Destination entity

Example:

- Visa paid via Noqodi:

- Source = Noqodi drawer
- Salary paid via bank:
 - Source = Bank drawer

System BLOCKS missing drawer.

5.6 Insufficient Funds Handling

Rule 5.6.1 — Hard Block

If drawer balance < required amount:

- ✕ Journal cannot be posted
- ✕ Payroll finalization blocked
- ▶ Flag “Insufficient Funds”

No overdraft unless explicitly allowed later.

5.7 Clearing & Suspense Accounts

Rule 5.7.1 — Temporary Holding

Used when:

- Platform salary expected but not received
- Fine paid but rider unknown
- Bank entry unclear

Rules:

- Must be cleared via reversal or settlement journal
 - Cannot remain uncleared beyond configurable days
-

5.8 Cashflow Reporting

System MUST show:

- Opening balance
- Inflows
- Outflows

- Closing balance
 - Per drawer
 - Per date range
-

5.9 HARD BLOCKS

System BLOCKS:

- Posting journal without drawer
- Editing drawer type after use
- Deleting drawers with history
- Posting if currency mismatch

STEP 6 — ACTION NEEDED / BLOCKING ENGINE

6.1 Core Philosophy

Rule 6.1.1 — No Silent Failures

- Any condition that:
 - Prevents correct accounting
 - Creates financial ambiguity
 - Risks cash loss
 - Approving PRO expenses by accountant

→ MUST surface as an **Action Needed item**

No background ignoring. No “we’ll fix later”.

6.2 Action Needed Object

Each Action Needed item has:

- ID
- Type
- Severity (Blocker / Warning / Info)
- Related Entity (Rider / Journal / Bike / Drawer)

- Reference ID
 - Created At
 - Resolved At
 - Resolution Notes
 - Responsible Role (Accountant / PRO / System)
-

6.3 BLOCKERS (System Stops Everything)

These **HARD BLOCK operations** until resolved.

6.3.1 Payroll Blockers

- Rider alias mismatch
- Duplicate payslip attempt
- Rider status = Retired
- Unassigned fines for that rider
- Insufficient drawer balance
- Salary sheet rider ID not found

→ Payroll finalization DISABLED

6.3.2 Accounting Blockers

- Journal imbalance
 - Missing receivable/payable flag
 - Journal without drawer
 - Currency mismatch
 - Attempt to edit posted journal
-

6.3.3 Fines & Bikes Blockers

- Fine with no bike plate match
 - Bike assignment overlap
 - Fine date outside any assignment
 - Multiple riders for same plate/time
-

6.4 WARNINGS (Allowed but Highlighted)

Warnings DO NOT block actions but:

- Must be acknowledged
- Are logged

Examples:

- Manual drawer adjustment
 - Large one-time deduction
 - Salary unusually low/high
 - Long uncleared clearing balance
-

6.5 INFORMATIONAL ALERTS

Pure visibility:

- Rider nearing zero balance
 - Loan almost recovered
 - Bike unused for X days
 - High PRO expense volume
-

6.6 Resolution Flow

Rule 6.6.1 — Explicit Resolution Required

Action Needed item:

- Cannot auto-disappear
- Requires:
 - User action
 - Or system event

Resolution records:

- Who resolved
 - When
 - How
-

Rule 6.6.2 — Auto-Resolution

Some items auto-resolve:

- Rider alias fixed
- Fine assigned
- Drawer funded

But still logged.

6.7 Role-Based Visibility

Role	Can View	Can Resolve
PRO	Own items	Limited
Accountant	All	All
Owner/Admin	All	All

6.8 UI Behavior (Important)

- Global **Action Needed counter**
- Red badge for blockers
- Click → direct deep link to problem
- Blocked buttons explain WHY

No guessing.

6.9 Escalation Rules

- Blocker unresolved > X days:
 - Escalates to Owner
 - Repeated same issue:
 - Flagged as pattern
-

6.10 HARD BLOCKS

System BLOCKS:

- Closing month with unresolved blockers
- Payroll run with active blockers
- Journal posting with unresolved system errors

STEP 7 — REPORTING & Accountant DASHBOARD

7.1 Core Philosophy

Rule 7.1.1 — Accountant Sees SYSTEM TRUTH, Not Filtered Truth

- Reports pull data **directly from ledger & journals**
- No dependency on:
 - UI state
 - User-entered balances
- Every number must be:
 - Traceable
 - Drill-down capable

If a number cannot be drilled down → it doesn't belong on dashboard.

7.2 Accountant Dashboard (Top Layer)

7.2.1 Financial Snapshot (At a Glance)

Accountant sees:

- Total Cash (sum of all drawers)
- Bank vs Cash vs Noqodi split
- Total Receivables (riders + others)
- Total Payables
- Net Position

All values:

- Real-time
- Clickable → ledger breakdown

7.2.2 Monthly Performance Panel

For selected month:

- Total salaries paid
- Total deductions recovered
- Net payroll outflow
- Total fines recovered
- Total visa / onboarding recovery

Used to quickly answer:

“Is this month healthy or leaking?”

7.3 Actor-Based Expense Reporting (Your Special Requirement)

Rule 7.3.1 — Expense by Initiator

Owner **MUST** be able to view:

- Expenses created by PRO
- Expenses created by Accountant
- System-generated expenses (Payroll)

Breakdown:

- By category
- By time range
- By drawer

This directly enforces **Rule 1.5.1**.

Rule 7.3.2 — Approval Funnel

Accountant can see:

- PRO-created → Approved
- PRO-created → Rejected
- Pending approvals (aging)

Detects:

- Laziness
 - Abuse
 - Bottlenecks
-

7.4 Rider-Level Reports

7.4.1 Rider Financial Summary

For any rider:

- Total salary earned
- Total deductions
- Outstanding balance
- Fine recovery status
- Loan recovery status

Includes:

- Timeline view
 - Category-wise totals
-

7.4.2 Historical Consistency

Reports MUST respect:

- Rider aliases
- Platform changes
- Past IDs

Accountant can pull:

“Rider X full financial history — regardless of company changes”

7.5 Payroll Reports

7.5.1 Payroll Register

For a given month:

- Rider
- Platform
- Gross salary
- Company deductions
- Net paid
- Drawer used

Exportable (Excel / PDF).

7.5.2 Deduction Aging Report

Shows:

- How long balances are outstanding
- Rider-wise
- Category-wise

Used to detect:

- Non-recoverable riders
 - Policy flaws
-

7.6 Fine & Bike Reports

7.6.1 Fine Recovery Report

- Total fines issued
 - Total recovered
 - Pending fines
 - Unassigned fines (RED FLAG)
-

7.6.2 Bike Utilization

Per bike:

- Assigned days
- Riders used
- Fines generated

Helps detect:

- Abuse
 - Poor asset usage
-

7.7 Drawer & Cashflow Reports

7.7.1 Drawer Statement

Per drawer:

- Opening balance
- All transactions
- Closing balance

Like a bank statement — audit-grade.

7.7.2 Cashflow Summary

- Inflows vs Outflows
 - Category-wise
 - Period-wise
-

7.8 Action Needed Summary (Accountant View)

Owner sees:

- Total blockers
- Blockers by category
- Aging of unresolved items
- Repeated patterns

This is **early warning system**.

7.9 Drill-Down & Traceability (MANDATORY)

Rule 7.9.1 — No Dead Numbers

Every number must allow:

- Click → journals
- Click → ledger entries
- Click → entity

If not drillable → remove it.

7.10 Export & Audit

Reports MUST:

- Export exactly what is shown
- Preserve totals
- Include:
 - Filters used
 - Date/time generated
 - Generated by user

Audit-safe.

STEP 8 — SECURITY, ROLES & PERMISSIONS

8.1 Core Security Philosophy

Rule 8.1.1 — Least Privilege

- Every user gets **minimum access required**
 - No shared accounts
 - No “temporary full access”
-

Rule 8.1.2 — All Actions Are Audited

Every critical action logs:

- User ID
- Role
- Timestamp

- Before & After state
- IP / Device (optional but recommended)

Logs are:

- Append-only
- Not editable
- Not deletable

8.2 System Roles (Hard Defined)

8.2.1 Accountant (Admin)

Capabilities:

- Create / approve journals
- Approve PRO expenses
- Run payroll
- Manage riders
- Assign bikes
- Resolve Action Needed
- Manage drawers

This role is **operational power**.

8.2.2 PRO (Operations)

Capabilities:

- Create riders (initial only)
- Upload documents
- Create expense drafts
- View assigned riders
- View own transaction history

Restrictions:

- ✗ Cannot approve
- ✗ Cannot post journals
- ✗ Cannot edit riders after creation
- ✗ Cannot see ledger totals

8.3 Permission Matrix (Critical)

Action	Accountant	PRO
Create Journal	✓	Draft only
Approve Journal	✓	✗
Reverse Journal	✓	✗
Run Payroll	✓	✗
Create Rider	✓	✓ (initial)
Edit Rider	✓	✗
Assign Bike	✓	✗
View Ledger	✓	Limited
View Own Expenses	Yes	✓

8.4 Data Visibility Rules

Rule 8.4.1 — PRO Visibility

PRO can see:

- Only journals he created
- Only riders he onboarded
- No totals
- No balances

This prevents financial leakage.

Rule 8.4.2 — Accountant Visibility

- Full operational view
 - Cannot see owner-only audit reports (optional)
-

8.5 Critical Action Protection

Rule 8.5.1 — Dual Confirmation

For high-risk actions:

- Manual drawer adjustment
- Journal reversal
- Payroll finalization

System requires:

- Confirmation dialog
 - Reason text
-

Rule 8.5.2 — Time Locks

Optional:

- Journals older than X days:
 - Can be reversed
 - But require owner note
-

8.6 Authentication

Rule 8.6.1 — Secure Login

- Username + Password
 - Password policy:
 - Min length
 - Complexity
 - Upper case, lower case, number and special character required.
-

8.7 Session & Access Control

- Auto logout after inactivity
 - Device-based login tracking
 - Optional IP whitelisting for accountant
-

8.8 Audit Log Scope

Logged events include:

- Login / logout
 - Journal creation / approval / reversal
 - Payroll run
 - Rider edits
 - Bike assignment
 - Drawer adjustment
 - Permission change
-

8.9 HARD BLOCKS

System BLOCKS:

- Role escalation without owner
- Deleting users with history
- Disabling audit logs
- Editing permissions inline

STEP 9 — END-TO-END SYSTEM WALKTHROUGH

SCENARIO OVERVIEW

- Rider: **Ali**
 - Platform: **Talabat** → later **Keeta**
 - Bike: **DXB-K-12345**
 - Visa cost: **2,000 AED (charged 5,000)**
 - Fine: **300 AED**
 - Salary: **January 2026**
 - Drawer: **Bank + Noqodi**
-

9.1 Rider Onboarding (PRO → Accountant)

9.1.1 PRO Creates Rider (Draft State)

PRO:

- Creates rider profile
- Enters:
 - Name
 - Emirates ID
 - Phone
- Uploads documents

System:

- Assigns `internal_rider_id`
 - Rider status = Active
 - PRO CANNOT edit later
-

9.1.2 Accountant Completes Rider Setup

Accountant:

- Reviews rider
- Adds:
 - Platform alias:
 - Talabat
 - External ID: TLB-9981
 - Valid from: 1 Jan 2026
- Assigns bike **DXB-K-12345**

System:

- Closes any previous alias (auto)
 - Assignment timestamped
-

9.2 Visa Expense (PRO → Accountant → Ledger)

9.2.1 PRO Adds Visa Expense

- Amount paid: 2,000
- Drawer: Noqodi
- Category: Visa
- Rider selected

System:

- Journal created (Unapproved)

- No ledger impact
 - Action Needed: Approval pending
-

9.2.2 Accountant Approves + Marks Receivable

Accountant:

- Approves journal
- Sets:
 - Receivable = 5,000
- Uses journal template

System:

- Journal posted
 - Ledger entries created:
 - Noqodi credited 2,000
 - Rider receivable debited 5,000
 - Drawer updated
-

9.3 Fine Occurs (Plate → Rider → Receivable)

9.3.1 Fine Entry

- Plate: DXB-K-12345
- Date: 15 Jan
- Amount: 300

System:

- Resolves bike → rider Ali
- Fine assigned
- Journal created:
 - Rider receivable +300

If assignment missing:

- Fine stays UNASSIGNED
 - Payroll blocked
-

9.4 Salary Sheet Upload (Talabat)

9.4.1 Accountant Uploads Sheet

Sheet row:

- External ID: TLB-9981
- Gross: 2,500

System:

- Resolves alias
 - Draft payslip created
-

9.5 Payroll Deduction Phase

Draft shows:

- Visa outstanding: 5,000
- Fine outstanding: 300

Accountant:

- Deducts:
 - Visa: 1,000
 - Fine: 300

System:

- Shows live remaining balances
-

9.6 Payroll Finalization (Point of No Return)

Accountant clicks **Finalize**.

System checks:

- No unassigned fines ✓
- Alias valid ✓
- Drawer sufficient ✓

System generates:

1. Salary payable journal
2. Deduction journals
3. Net payment journal

Drawer:

- Bank credited (outflow)

Payslip locked.

9.7 Platform Switch (Talabat → Keeta)

9.7.1 Alias Change

Accountant:

- Closes Talabat alias (31 Jan)
- Creates Keeta alias (1 Feb)
- New external ID: KTA-4412

Internal rider ID unchanged.

9.8 Historical Payroll Integrity

Owner pulls:

“Ali salary Jan 2026”

System:

- Uses Talabat alias
- Shows correct history

Feb salary:

- Uses Keeta alias

No conflict.

9.10 Failure Scenario (System Protects)

Unassigned Fine Exists

- Payroll button disabled
- Action Needed visible
- Cannot bypass

FINAL VERDICT

- ✓ Accounting-correct
- ✓ Payroll-safe
- ✓ Rider-proof
- ✓ Audit-grade
- ✓ Scalable
- ✓ Developer-ready